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R Notebooks: ECG Heart rate and HRV Processing

The heart rate, which is calculated from Electrocardiography (ECG) or photoplethysmogram (PPG) signal, is coupled with autonomic nervous system activity. It provides a better understanding of physiological arousal and a suitable proxy for examining how we feel. Heart rate variability (HRV) is associated with emotional regulation, social skills, and cognitive processes. Note that HRV is calculated from ECG signals (not from PPG signals, since PPG signals are not recommended for HRV analysis).

This article describes the ECG signal peak detection and calculations of Heart Rate (HR) and HRV based on our new R-notebook integration.

The R-notebook is called: Heart Rate and Heart Rate Variability (HRV). It computes the single respondent peak detection, calculations of HR and HRV (SDNN and RMSSD) during each stimulus and from iMotions 10.0.4 automated event markers. From iMotions 10.0.10 onwards, frequency HRV metrics are available. Based on analysis of the power of different frequency ranges of the Inter-Beat-Intervals of ECG, frequency HRV metrics are able to give more insights into the sympathetic nervous system (SNS) and parasympathetic nervous system (PNS) balance that can be used to answer a lot of research questions regarding mental workload, drowsiness, alertness detection etc.

Supported sensors:

- Shimmer ECG
- Biopac ECG (ECG100C) through MP150 and MP160
- Biopac Bionomadix ECG
- Bluetooth Low Energy (BLE) devices such as Polar Belt H10
- Biosignal Plux ECG (from iMotions 10.0.7 onwards)
- ABM B-Alert ECG channel

Regarding what is Bluetooth Low Energy sensors and how to connect them in iMotions, please refer to this article.

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Devices	ECG Related Raw Signals You Could Obtain From the Device	Metrics/Signals You Could Obtain from the Notebook
Shimmer ECG	ECG, Heart Rate ECG (LL-RA), IBI ECG (LL-RA)	Metrics (Percentage of missed peaks (%), Average Heart Rate, average IBI, SDNN, RMSSD, VLF Power, LF Power, HF Power, LF/HF (%), Normalized LF Power, Normalized HF Power, Relative LF Power (%), Relative HF Power (%)), Signals (R Peaks Occurrences, momentary heart rate, momentary HRV)
Biopac ECG100C	ECG	Metrics (Percentage of missed peaks (%), Average Heart Rate, average IBI, SDNN, RMSSD, VLF Power, LF Power, HF Power, LF/HF (%), Normalized LF Power, Normalized HF Power, Relative LF Power (%), Relative HF Power (%)), Signals (R Peaks Occurrences, momentary heart rate, momentary HRV)
BLE sensors with R-R intervals like Polar Belt H10	Heart rate, R-R intervals	Metrics (Average Heart Rate, the average IBI, SDNN, RMSSD, VLF Power, LF Power, HF Power, LF/HF (%), Normalized LF Power, Normalized HF Power, Relative LF Power (%), Relative HF Power (%) will be calculated). Signals (momentary HRV)
BLE sensors without R-R intervals like Polar Verity Sense	Heart rate	Metrics (Average Heart Rate will be calculated)

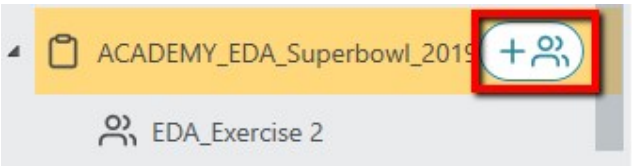
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Biosignal Plux ECG	ECG	Metrics (Percentage of missed peaks (%), Average Heart Rate, average IBI, SDNN, RMSSD, VLF Power, LF Power, HF Power, LF/HF (%), Normalized LF Power, Normalized HF Power, Relative LF Power (%), Relative HF Power (%)), Signals (R Peaks Occurrences, momentary heart rate, momentary HRV)
ABM B-Alert	ECG	Metrics (Percentage of missed peaks (%), Average Heart Rate, average IBI, SDNN, RMSSD, VLF Power, LF Power, HF Power, LF/HF (%), Normalized LF Power, Normalized HF Power, Relative LF Power (%), Relative HF Power (%)), Signals (R Peaks Occurrences, momentary heart rate, momentary HRV)

1. How to Analyze in iMotions Desktop

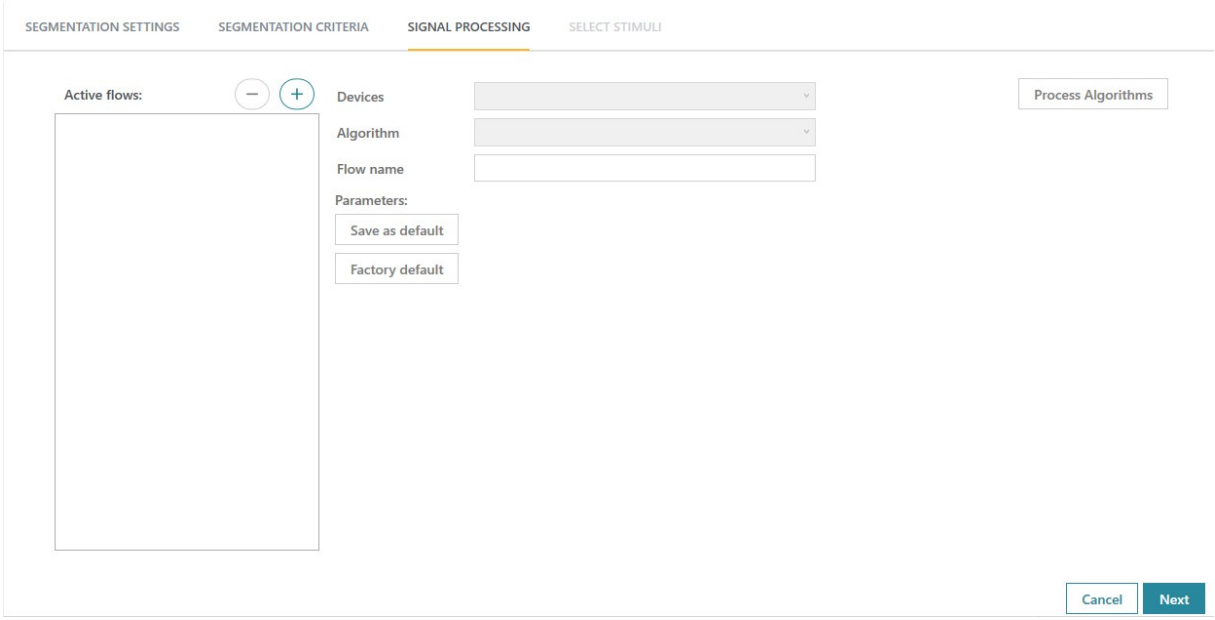
As for the Heart Rate and Heart Rate Variability R Notebook, the following steps need to be followed:

1. Create a new analysis for the study for which you wish to run the R-Notebook:



2. Advance through the steps "Segmentation settings" with your preferred selections and "Segmentation criteria" until you reach the step "Signal processing".

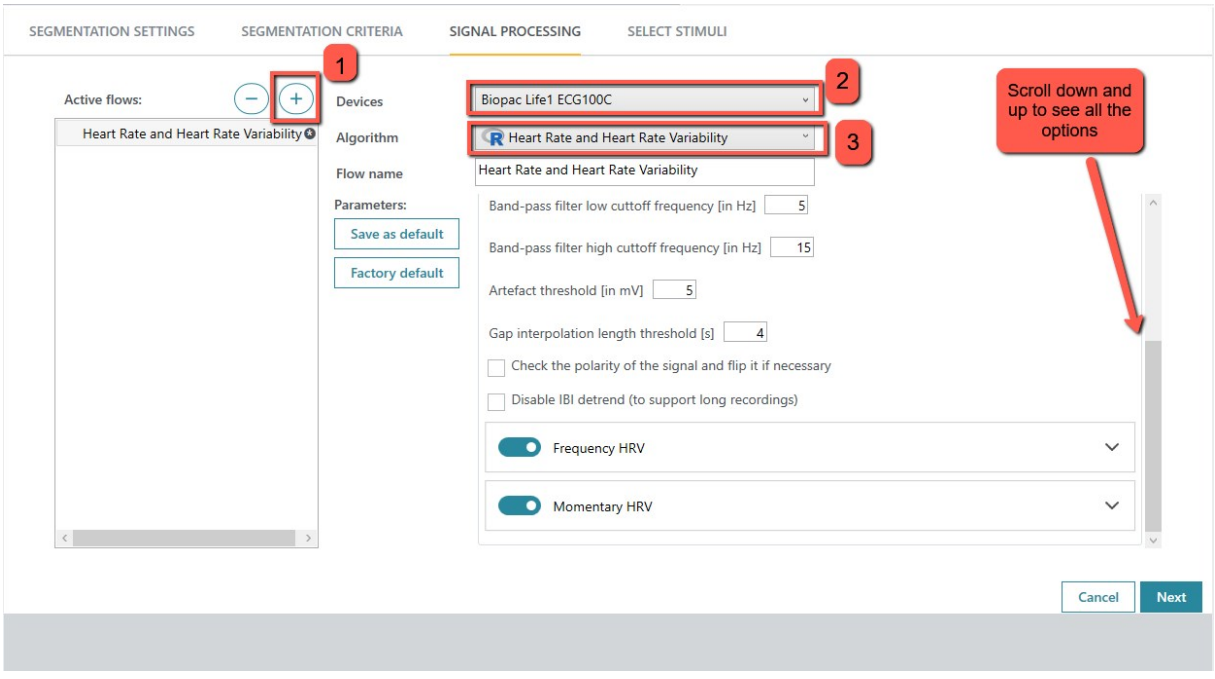
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3. As shown in the screenshot below, by clicking the "+" button next to "Active flows", you can select the algorithms you want to use for signal processing. Select the sensor for which you want to apply the ECG R-Notebook in the drop-down "Device" menu. Select the algorithms in the drop-down “Algorithm” menu and name the analysis in the "Flow name" menu.

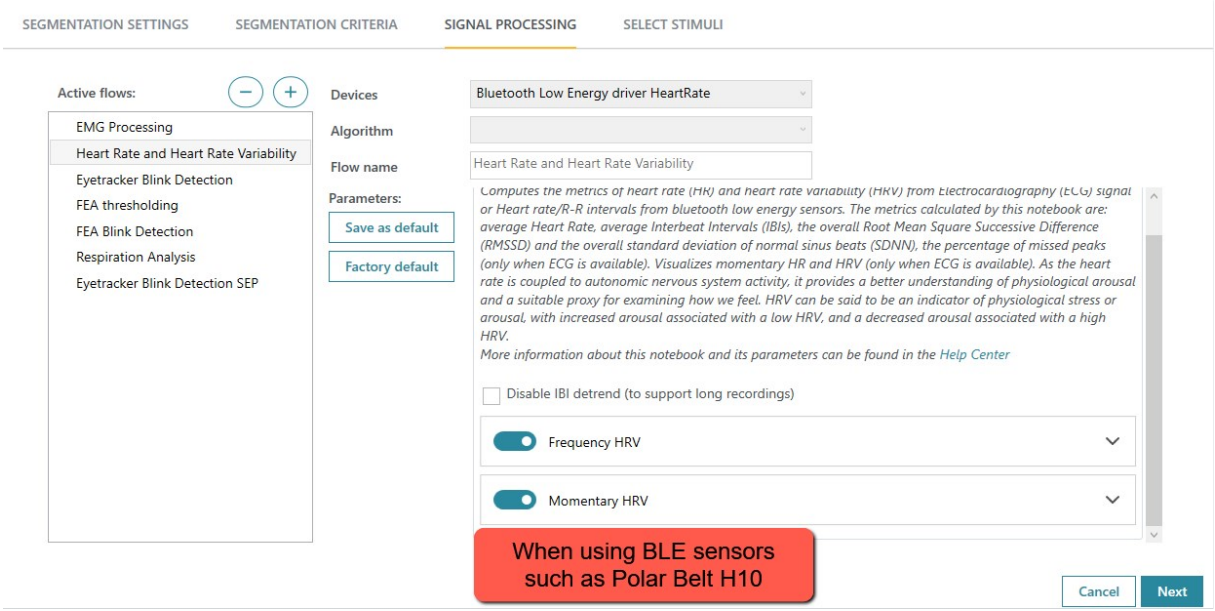
Alter the parameters to suit your research needs, then click on “Next”. The parameters will be described in detail in the next section of this article: Algorithm Introduction.

Note: it is possible now (from version 8.1 and onwards), to run the notebook on multiple sensors of the same kind. E.g., you can run the notebook on two Shimmer devices here by adding two Active flows with the devices you would like to analyze on.

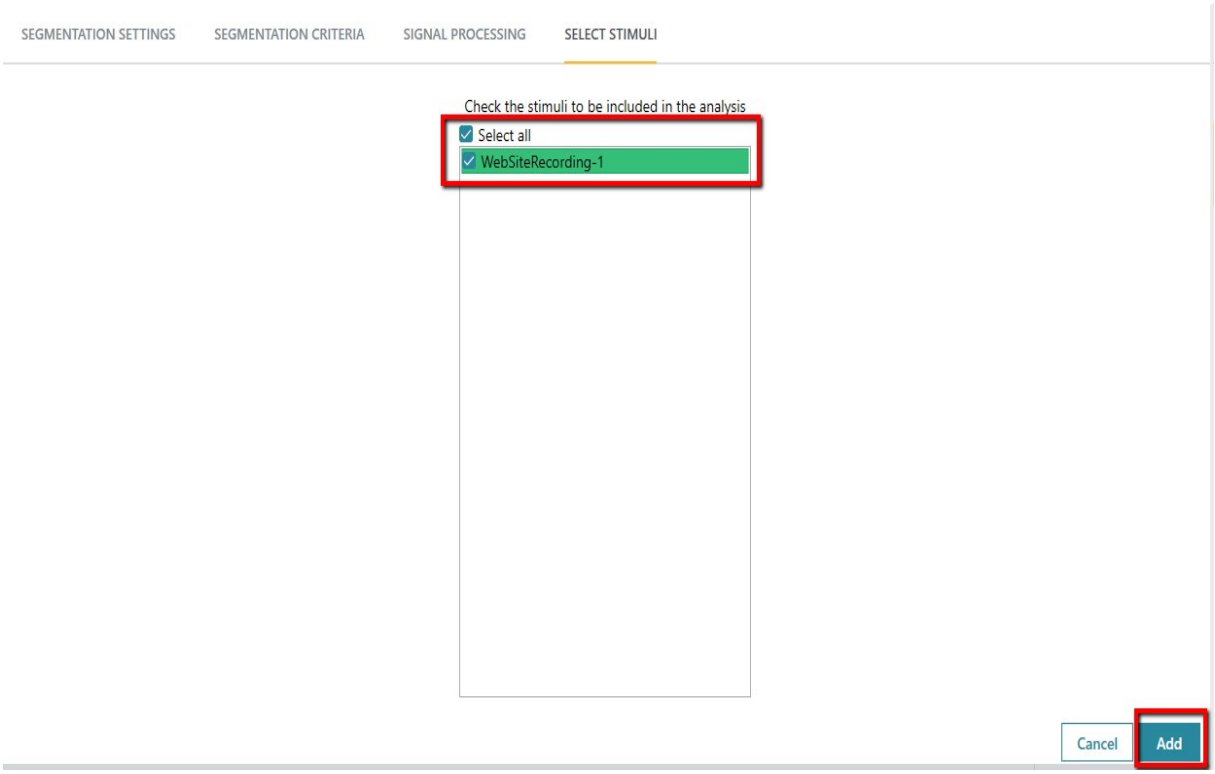


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For BLE sensors, you can also choose whether to “Disable IBI detrend (to support long recordings)” or not as shown in the screenshot below. Due to the signal properties from BLE sensors, peak detection calculation will be skipped. The algorithm will be explained more in the section Algorithm Introduction.



5. Select the stimulus to be included in the analysis. Then click the "Add" button to add the analysis and close the dialog.



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2. Algorithm Introduction

(1) ECG Peak Detection

for ECG signal from Shimmer, Biopac, Biopac Bionomadix, ABM B-Alert ECG channel, Biosignal Plux that stream in raw ECG signals:

1. *Shimmer ECG ONLY*: by defining parameter [Shimmer ECG channel name]: "ECG LL-RA CAL" or “ECG LA-RA CAL”, you can choose which channel you want to apply the R-Notebook on.
2. The signal is first bandpassed to reduce the influence of muscle noise, powerline noise (US 60 Hz ; Europe 50 Hz), baseline wander, and T-wave interference. A classic Butterworth band-pass filter with a total order of 4 is applied. The desirable passband to maximize the QRS energy is approximately 5-15 Hz (Goovaerts et al. 1976),(Thakor, Webster, and Tompkins 1983). The passband used in this analysis is between the [Band-pass filter low cuttoff frequency [in Hz]] and [Band-pass filter high cuttoff frequency [in Hz]] (default value 5 - 15 HZ). The filtering process was performed in forward and reverse direction. Data points with an absolute value superior to the selected Artefact threshold [in mV] (default 5 mV) are considered outliers and removed for further analysis. The filtered signal is then normalized in arbitrary units between 0 and 1000.
3. [Gap interpolation length threshold [ms]] is used to identify possible gaps in the signal due to signal drops. If a gap in the signal is shorter than this threshold, then the missing data is linearly interpolated. Otherwise, the gap is left and each of the signal fragments is processed separately.
4. Detection of possible peaks: a squared double difference method is applied to the filtered signal (Sadhukhan and Mitra 2012). The difference signal is compared with a moving maximum in a window of length 4 seconds. A possible peak is marked every time the difference signal crosses above 3% of the moving maximum.

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5. The adaptive algorithm runs through the possible peaks to identify the real peaks: we use a moving window of 6 seconds (advancing with an overlap). A local linear model is fit to the 3 largest ECG values among the possible peaks in the window. The linear model is then varied with a coefficient to lowering its intercept between 0% to 50%. For each value of the coefficient, we compute a trial in which we select the peaks that lie above the linear model threshold, and for each trial we compute the local heart rate and The Root Mean Square of Successive Differences (RMSSD) values in the window. Among all the trials, we select the solution with the lowest RMSSD among those with a heart rate value within 30% of the heart rate value in the previous valid time window.
6. Once the peaks are selected in each moving window, we recompute the Inter-Beat Interval (IBI) and Heart Rate values for the selected ones. We mark as “missed peaks” the timestamps where the IBI signal becomes larger than 1.4 times a running median of itself over the adjacent 19 peaks. Incorrect IBIs are removed before metrics calculation.
7. ECG polarity flipping is possible now if the checkbox [Check the polarity of the signal and flip it if necessary] is checked. If the checkbox is checked, the sign of the signal will be automatically checked and the signal will be flipped if polarity detected to be reversed.
8. [Disable IBI detrend (to support long recordings)]

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9. To calculate the frequency domain metrics the conversion of time domain into the frequency domain is needed, which is done through the calculation of the power spectral density (PSD). This is based on the IBI detrended signal, with the same method used for the time domain, but with a sampling frequency of 4 Hz for the interpolation (Heine et al. 2017). The Welch method, which uses the Fast-Fourier Transformation (FFT), is used for the estimation of PSD. For the window overlapping the method Hann window is applied with a window width of 64 s (the same as 256 sample points) and with 50% overlap between neighboring windows. The PSD from all windows are then averaged to get the PSD signal. This method has been selected to create a more robust estimation of the PSD signal. The PSD signal is divided into three frequency intervals; Very low frequencies [in Hz] (VLF, default span from 0.0033 - 0.04 Hz), Low frequencies [in Hz] (LF, default span from 0.04 - 0.15 Hz) and High frequencies [in Hz] (HF, default span from 0.15 - 0.4 Hz), where the spectral power is calculated as the area under the curve (AUC) of each interval. If the Compute normalized frequency metrics are checked off, the normalized frequency metrics is calculated by taking either high or low frequency and divide it with the sum of high and low frequency and multiply it with 100. The relative frequency metrics is calculated by taking either high or low frequency and divide it with the total power (the sum of all frequencies: TP = HF + LF + VLF) and multiply it with 100 (Shaffer and Ginsberg 2017).

> **Note**
>
> *Please ensure a duration of at least 64 seconds for the notebook to deliver any frequency metrics data.*

for ECG signal from BLE sensors that stream in R-R intervals signals:

1. The Inter-Beat Interval (IBI) signal (same as R-R interval) is directly retrieved from the raw data. The heart rate signal is then calculated based on the IBI signal: heart_rate=1000* 60/IBI_signal

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2. To calculate the time domain metrics (RMSSD (ms) and SDNN (ms)): The IBIs signals are first detrended to obtain stationary signals. An advanced detrending procedure based on smoothness priors approach is applied (Tarvainen, Ranta-aho, and Karjalainen Feb./2002). The smoothing parameter lambda is set as 500, which equals a cutoff frequency of 0.035 Hz. The method operates like a time-varying highpass filter, smoothing the data according to the value of the smoothing parameter lambda. The bigger the smoothing parameter is, the lower the cutoff frequency of the filter will be.
3. To calculate the frequency domain metrics the conversion of time domain into the frequency domain is needed, which is done through the calculation of the power spectral density (PSD). This is based on the IBI detrended signal, with the same method used for the time domain, but with a sampling frequency of 4 Hz for the interpolation (Heine et al. 2017). The Welch method, which uses the Fast-Fourier Transformation (FFT), is used for the estimation of PSD. For the window overlapping the method Hann window is applied with a window width of 64 s (the same as 256 sample points) and with 50% overlap between neighboring windows. The PSD from all windows are then averaged to get the PSD signal. This method has been selected to create a more robust estimation of the PSD signal. The PSD signal is divided into three frequency intervals; Very low frequencies [in Hz] (VLF, default span from 0.0033 - 0.04 Hz), Low frequencies [in Hz] (LF, default span from 0.04 - 0.15 Hz) and High frequencies [in Hz] (HF, default span from 0.15 - 0.4 Hz), where the spectral power is calculated as the area under the curve (AUC) of each interval. If the Compute normalized frequency metrics is checked off, the normalized frequency metrics is calculated by taking either high or low frequency and divide it with the sum of high and low frequency and multiply it with 100. The relative frequency metrics is calculated by taking either high or low frequency and divide it with the total power (the sum of all frequencies: TP = HF + LF + VLF) and multiply it with 100 (Shaffer and Ginsberg 2017).

> **Note**
>
> *Please ensure a duration of at least 64 seconds for the notebook to deliver any frequency metrics data.*

for ECG signal from BLE sensors that stream in only Heart rate signals:

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The Heart Rate metrics is calculated as the overall mean of the 'Heart rate' signal. The HRV metrics will not be calculated due to the missing of R-R intervals.

(2) Momentary HRV Calculation

1. To calculate the momentary HRV calculation: the length of the available ECG signal is first checked to see whether it is longer than [Momentary HRV moving window length [s]] (300 seconds (5 min) as default). The following calculations will only be conducted if this condition is fulfilled.
    a. the raw IBI time-series from peak detection is first interpolated and downsampled to 4 Hz.
    b. The IBI time-series is then detrended to obtain stationary signals. An advanced detrending procedure based on smoothness priors approach is applied (Tarvainen, Ranta-aho, and Karjalainen Feb./2002). The smoothing parameter lambda is set as 500, which equals a cutoff frequency of 0.035 Hz.
    c. The adaptive algorithm runs through the interpolated and detrended IBI time-series to calculate the variability time-series: a moving window of [Momentary HRV moving window length [s]] (300 seconds (5 min) as default) advancing with a time step of [Momentary HRV window increment step [s]] (60 seconds (1 min) as default) is used. For each window, the SDNN or the RMSSD is calculated. You can choose whether to run the SDNN or the RMSSD when scrolling down in that section of the software [Momentary HRV calculation method].

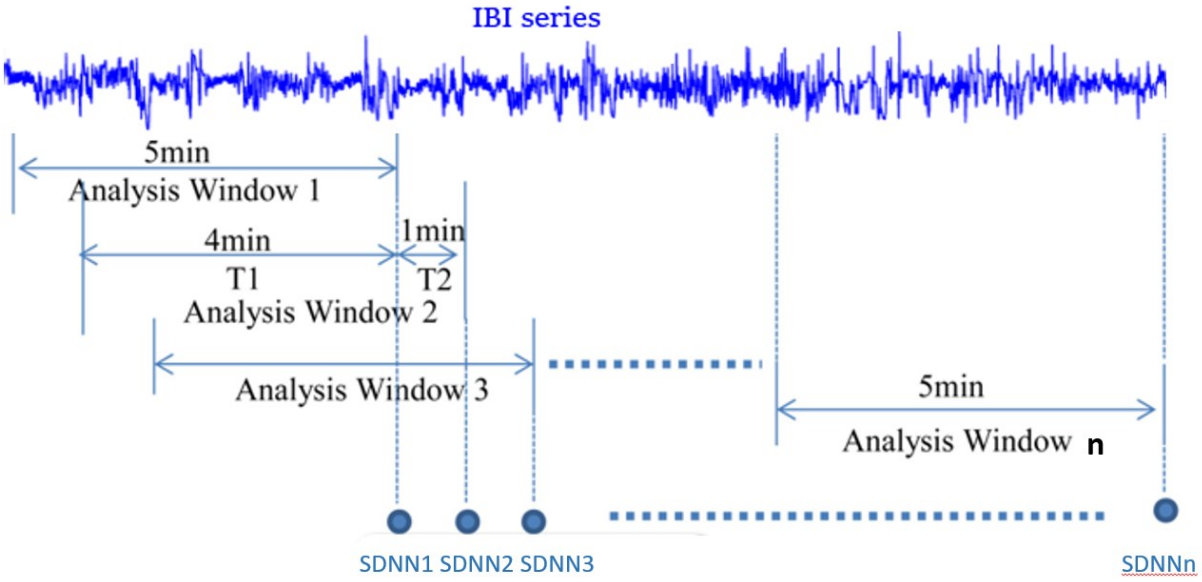
> Note
> *A moving window of at least 5 minutes is recommended for short-term HRV analysis (Heart J 1996) and has been applied for capturing momentary HRV with a recommended increment step of 1 minute (Zhang et al. 2015). To make use of all the data, the increment step of each moving window is recommended to be shorter or equal to the moving window length.*

Once the variability time-series is generated, a corresponding step plot is generated: x axis denotes the time spent since the stimulus starts, y axis denotes the calculated momentary HRV.

Momentary HRV algorithm schematic graph:

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Example: [Momentary HRV moving window length [s]] 300 seconds (5 min) and [Momentary HRV window increment step [s]] 60 seconds (1 min) are used in this example to calculate momentary SDNN:



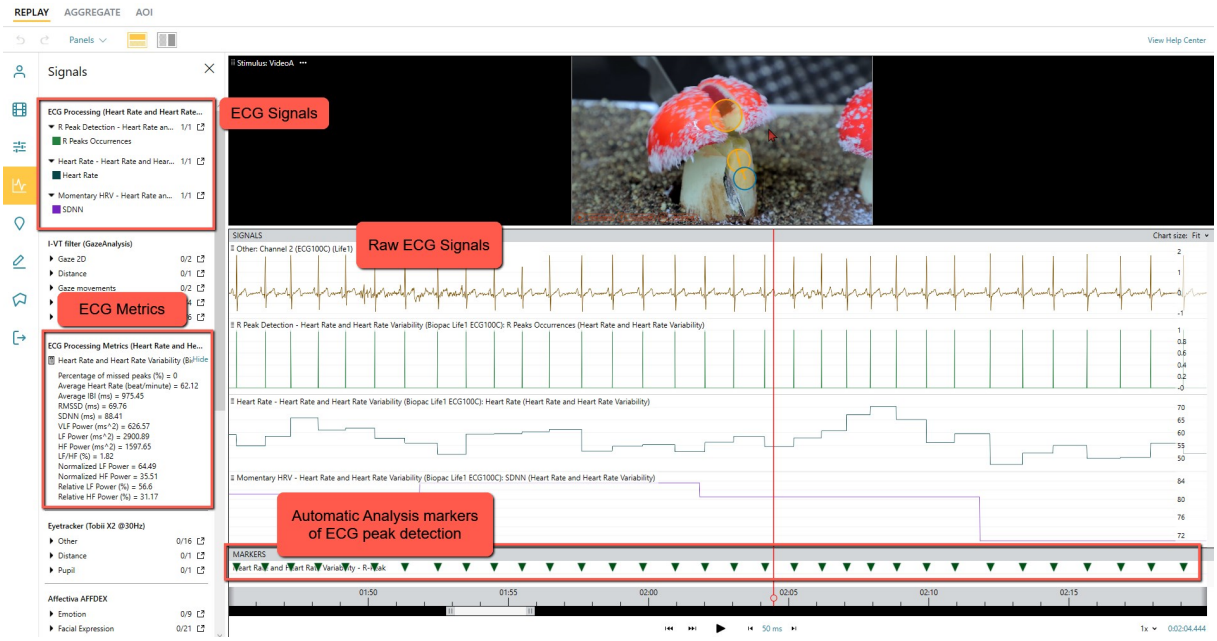
3. The Outputs

(1) Desktop replay:

If you are analyzing ECG signals from Shimmer, Biopac, Biopac Bionomadix, Biosignal Plux that stream in raw ECG signals, or BLE sensors that stream in R-R intervals signals like Polar Belt H10:

In the individual replays of data, the detected peaks and calculated metrics/signals are displayed as follows:

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- ECG Metrics:
  - [Percentage of missed peaks (%)]: the percentage of missed peaks out of all detected peaks. Please note that the metric's denominator is only taking into account the peaks in between the *first detected* peak and the *last detected* peak – therefore, if the first/last (a few) peaks are missed, all peaks in between though are detected, the metric will still show 0%.

> **Notes**
>
> Please note that this metrics is not computed for BLE sensors like Polar Belt H10.

  - [Average Heart Rate (beat/minute)]: the average beat per minute the respondent had during this stimulus/scene. Calculated from raw IBI time-series before any interpolation and detrending.
  - [Average IBI (ms)]: the average inter-beat intervals the respondent had during this stimulus/scene. In the unit of millisecond. Calculated from raw IBI time-series before any interpolation and detrending.
  - [RMSSD (ms)]: the overall root mean square of successive differences between normal heartbeats the respondent had during this whole stimulus/scene. In the unit of millisecond. Calculated from interpolated and detrended IBI time-series.

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- [SDNN (ms)]: the overall standard deviation of IBIs the respondent had during this whole stimulus/scene. In the unit of millisecond. Calculated from interpolated and detrended IBI time-series.
- [VLF Power (ms²)]: Absolute power of the very-low-frequency band, of the IBIs the respondent had during this whole stimulus/scene.
- [LF Power (ms²)]: Absolute power of the low-frequency band, of the IBIs the respondent had during this whole stimulus/scene.
- [HF Power (ms²)]: Absolute power of the high-frequency band, of the IBIs the respondent had during this whole stimulus/scene.
- [LF/HF]: Ratio of LF-to-HF power, of the IBIs the respondent had during this whole stimulus/scene.
- [Normalized LF Power (%)]: Normalized power of the low-frequency band, of the IBIs the respondent had during this whole stimulus/scene. It is computed by taking low frequency and dividing it with the sum of high and low frequency and multiply it with 100. Simplified as LF Power (ms²)/(LF Power (ms²) + HF Power (ms²)) * 100%
- [Normalized HF Power (%)]: Normalized power of the high-frequency band, of the IBIs the respondent had during this whole stimulus/scene. It is computed by taking high frequency and dividing it with the sum of high and low frequency and multiply it with 100. Simplified as HF Power (ms²)/(LF Power (ms²) + HF Power (ms²)) * 100%
- [Relative LF Power (%)]: Relative power of the low-frequency band, of the IBIs the respondent had during this whole stimulus/scene. It is calculated by taking low frequency and divide it with the total power (the sum of all frequencies: TP = HF + LF + VLF) and multiply it with 100%. Simplified as LF Power (ms²)/(total power (ms²)) * 100%
- [Relative HF Power (%)]: Relative power of the high-frequency band, of the IBIs the respondent had during this whole stimulus/scene. It is calculated by taking high frequency and divide it with the total power (the sum of all frequencies: TP = HF + LF + VLF) and multiply it with 100%. HF Power (ms²)/(total power (ms²)) * 100%

- ECG Signals:

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- [R Peak Detection]: R Peaks Occurrences. ECG peak samples (ECG values at the peak) are assigned a value of "1". Other samples outside the peak are assigned a value of "0". The signal is directly from peak detection results before any interpolation.

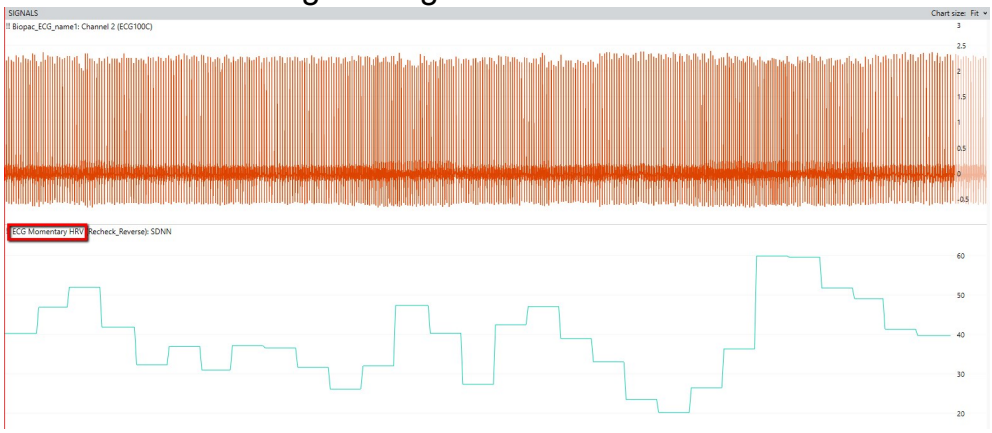


> **Notes:**
>
> Please note that this signal is not computed for BLE sensors like Polar Belt H10.

- [Heart Rate]: the running heart rate with each step value calculated from the corresponding IBI of two consecutive detected peaks.

> **Notes:**
>
> Please note that this metrics is not computed for BLE sensors like Polar Belt H10.

- [Momentary HRV]: the running momentary HRV signal with a step duration of [Momentary HRV window increment step [s]] represents the HRV values calculated from the moving window of [Momentary HRV moving window length [s]]. There can be an absence of signals near the end of the stimulus due to insufficient signal length.



If you are analyzing BLE sensors that ONLY stream in Heart Rate signals:

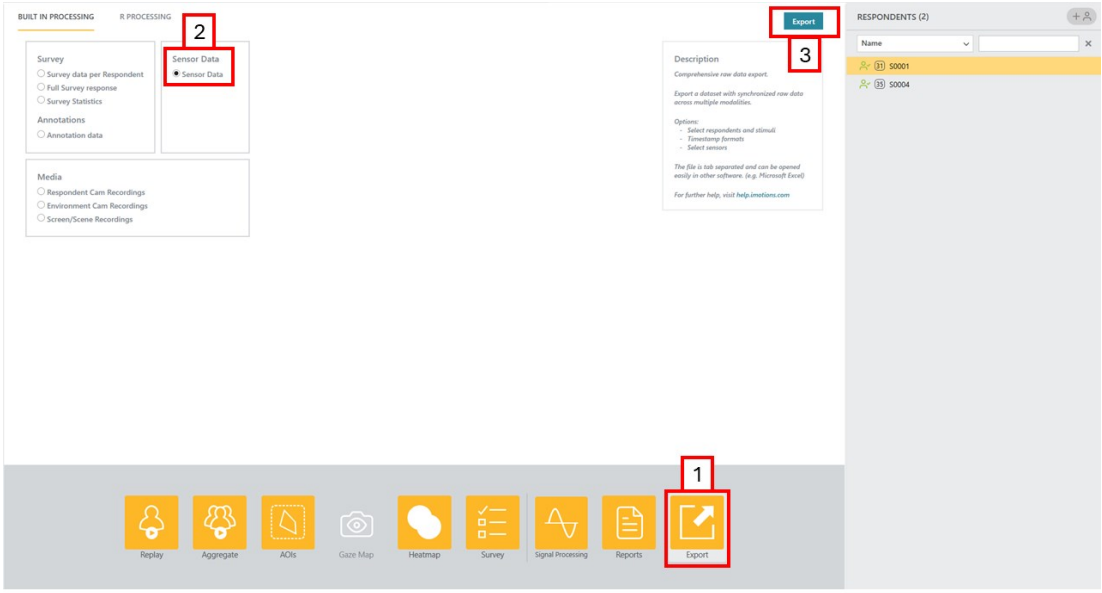
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- ECG Metrics: Average Heart Rate (beat/minute) will be calculated and displayed.
- ECG Signals: There will be no calculation or display of signals for this type of BLE sensor in the desktop replay.

(2) Exported files:

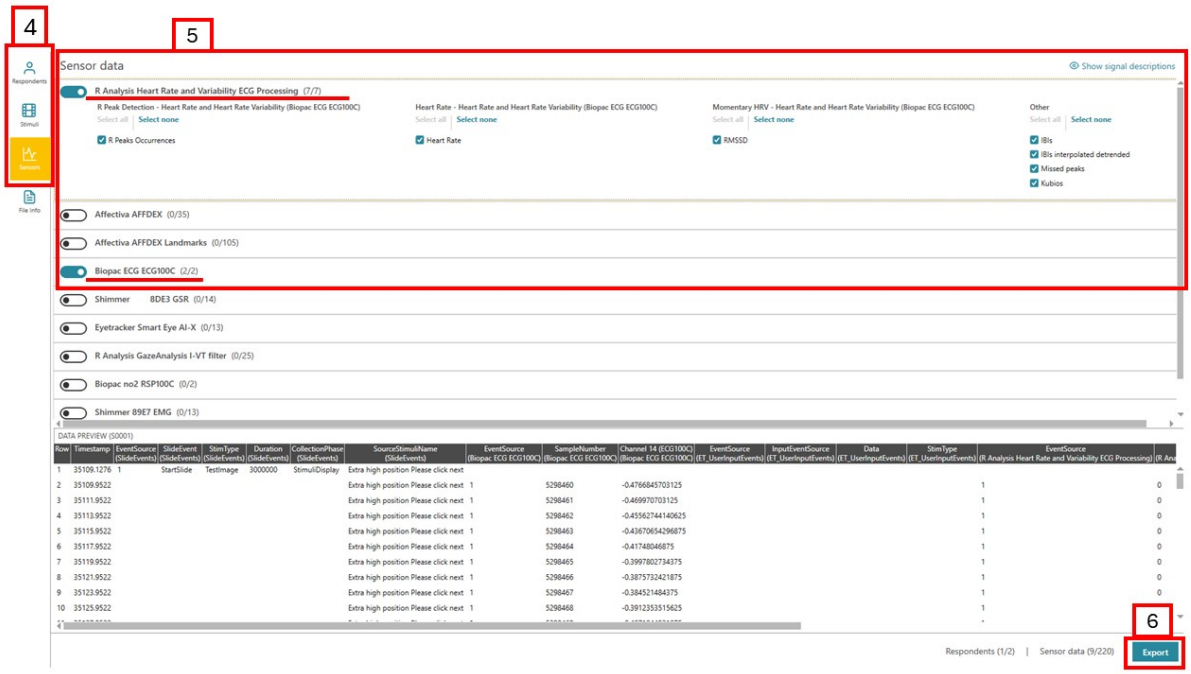
There are three types of files you can access: Sensor Data file, and 2 files from R processing (ECGMetrics and ECGPeakTable).

- Sensor Data file: raw sensor data export. It can also include R-processed signals by ticking the "R analysis Heart rate and Heart rate variability processing" as shown
  in step 5, as shown in the screenshots below. For iMotions 10.2 and earlier iMotions versions, to do that you will need to select the “Processed and imported data" option instead.



and then:

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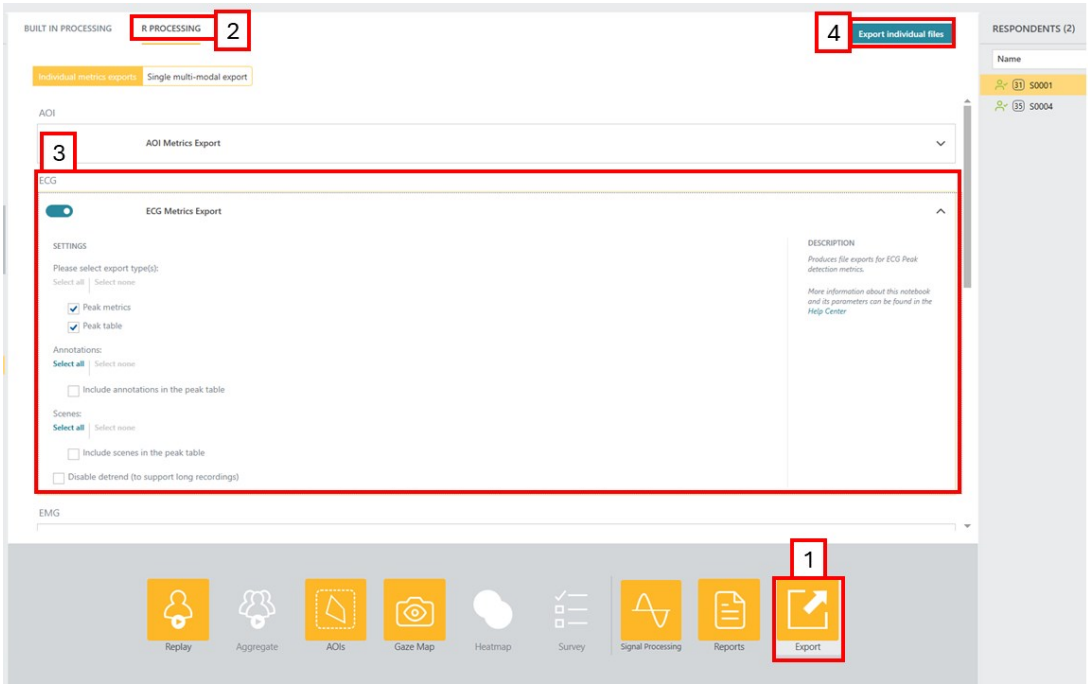


- ECGMetrics file: Processed results from R-Notebooks across respondents and stimuli. You can access the exported file by (1) clicking on the 'Export' button, (2) under 'R PROCESSING', you can (3) select the 'Peak metrics' exports , and then (4) click on 'Export' button to generate the file.

- ECGPeakTable file: from iMotions 10.4 and onwards. To get this file, select the ‘Peak table’ option as shown in the screenshot below. This export file specifies the Peak index, Peak time and inter-beat interval of every ECG peak (QRS segment) of every stimulus and, if you select the “include annotations in the peak table” option, of every annotation as well.

Please refer to *this article* for a detailed description of all the exported files.

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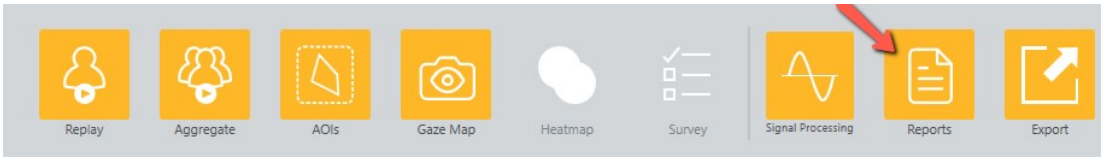


(3) Reports:

A one-page readout is generated, along with presenting the actual R Code used to calculate the results, descriptions of each of the analysis steps, and links to the articles that were referred to when writing the algorithm's script.

In brief, to find the Reports file:

1. Click the Reports button shown in the screenshot in the section above
2. As a result of that, you will be able to open the HTML report for every stimulus of your choice to see the R-Notebook readout.



In the opened Report, one can also find the overview of R Notebook results in both plots and metrics. For Heart Rate and Heart Rate Variability R Notebook, there will also be a spectrogram of the signal that describes the power spectral density (PSD) present in the signal as a function of frequency (as shown below).

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ECG Signal

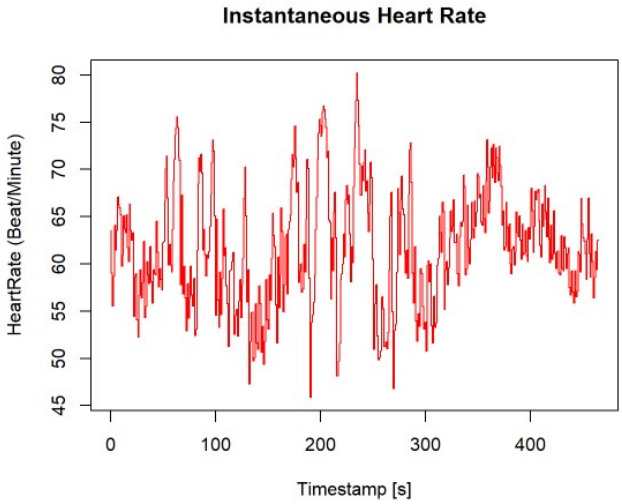
VideoA Hilary

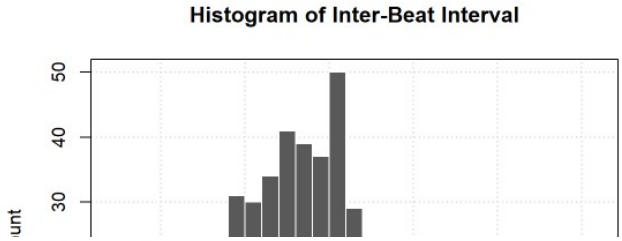
Easily switch between stimulus here

Show

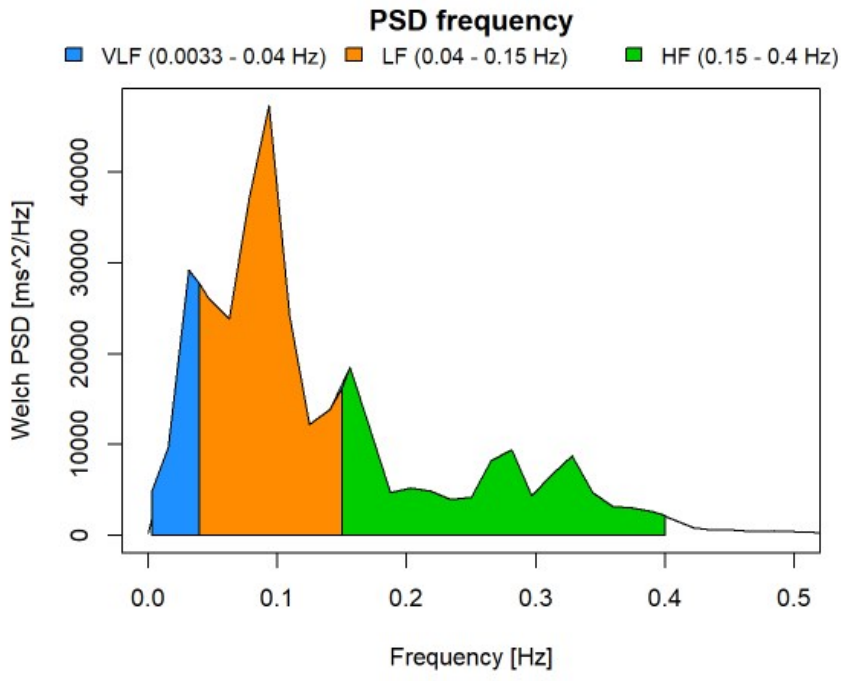
Note: The R-peak detection algorithm estimates that 0 % of the peaks are missed.

Time domain



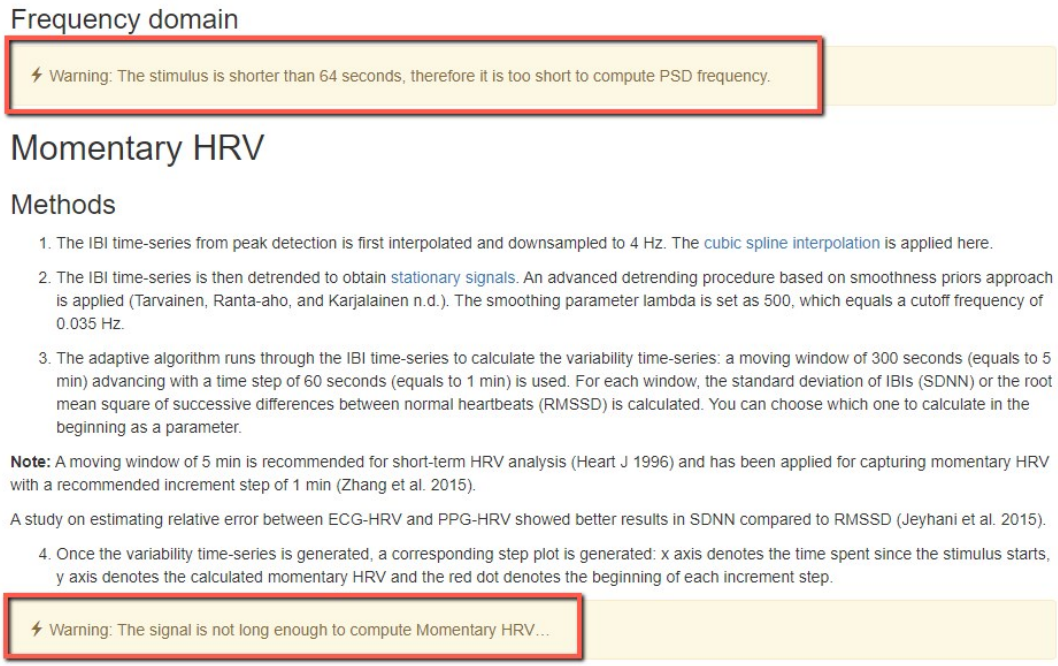


Frequency domain



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One can also use Reports to find troubleshooting information. For example, as shown below, the R Notebook can not run due to too short stimulus duration. This type of information will be provided in the Report.



References

Goovaerts, Henk G., Henk H. Ros, Van Den AkkerThomas J., and Hans Schneider. 1976. “A Digital QRS Detector Based on the Principle of Contour Limiting.” *IEEE Transactions on Biomedical Engineering* BME-23 (2): 154–60. https://doi.org/10.1109/TBME.1976.324577.

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